# Machine-learning surrogates for molecular hydrogen density in three-dimensional interstellar medium simulations using multi-scale spatial features

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**ABSTRACT**

Computing the molecular hydrogen number density $n_{H_2}$ self-consistently in three-dimensional interstellar-medium (ISM) simulations requires solving stiff non-equilibrium chemistry coupled to radiative transfer, a cost that limits the resolution and parameter coverage of modern calculations. We evaluate machine-learning surrogates that predict $\log_{10}(n_{H_2})$ cell-by-cell on a controlled suite of seven $128^3$ UV-only chemistry simulations spanning $G_0 \in \{0.1, \ldots, 6.4\}$ Habing (14.7 million cells), under leave-one-$G_0$-out cross-validation: each fold withholds an entire simulation at a UV field strength absent from training. Augmenting 15 per-cell physical features with 45 multi-scale box-filter neighbourhood averages (kernel widths 3, 5, 7 voxels; computed in ≈30 s by separable convolution) raises the mean log-space $R^2$ of gradient-boosted trees from 0.950 to 0.963, and a Ridge-stacked ensemble with a wide multilayer perceptron reaches $R^2 = 0.988 \pm 0.013$. Because one input, the $H_2$ self-shielding factor $f_{H_2}$, is a non-local quantity produced by the chemistry solver, we quantify its contribution with an ablation: without $f_{H_2}$, the pointwise models retain $R^2 = 0.909$–$0.911$, the spatial features recover roughly half of the loss, and the fully solver-independent stacked ensemble retains $R^2 = 0.977$. A 3D U-Net baseline trained on the same suite attains $R^2 = 0.974 \pm 0.035$ at its best configuration, matching the tabular models on interpolation folds but degrading at the extrapolation boundaries and varying strongly across training configurations, at roughly two orders of magnitude higher training cost. Within-cube sampling experiments show that reconstruction quality is controlled by sampling geometry rather than sample volume: 10 per cent uniformly random coverage yields $R^2 = 0.977$ while a contiguous half-cube fails entirely.

**Key words:** ISM: molecules – ISM: clouds – astrochemistry – methods: numerical – methods: statistical

## 1 INTRODUCTION

The formation of molecular hydrogen on dust-grain surfaces (Cazaux & Tielens 2004) and its photodissociation by Lyman–Werner UV photons (Habing 1968; Draine & Bertoldi 1996) together determine the molecular content of the interstellar medium (ISM). Because $H_2$ is both an important low-temperature coolant and the direct precursor of CO and the other molecular tracers used to infer star-formation rates (Narayanan et al. 2012), the $H_2$ abundance sets the initial conditions for the star-formation sequence. Computing it self-consistently within a three-dimensional hydrodynamics simulation requires integrating a stiff system of non-equilibrium chemistry equations at every grid cell, tracking the competition between $H_2$ formation on grain surfaces, UV photodissociation in the Lyman–Werner band (11.2–13.6 eV), and the $H_2$ self-shielding that occurs when $H_2$ columns attenuate their own dissociating radiation (Wolcott-Green et al. 2011). In simulations with $10^6$–$10^7$ cells this chemistry solver can dominate the compute budget, and covering a wide UV-parameter range at full resolution is often infeasible (Grassi et al. 2014).

Two families of shortcuts exist. The older one replaces the solver with analytic or fitted formulae for the atomic-to-molecular transition, derived from photodissociation-region theory or calibrated on simulations (Krumholz et al. 2009; Gnedin & Kravtsov 2011; Sternberg et al. 2014; Bialy & Sternberg 2016); these are fast and interpretable, but each is tied to the equilibrium assumptions, geometry, and averaging scale for which it was derived. The newer family trains machine-learning (ML) surrogates on solver outputs: de Mijolla et al. (2019) emulate UCLCHEM abundances for molecular-line modelling, Holdship et al. (2021) advance a thermochemical state through a time-step with the Chemulator network, Grassi et al. (2022) compress a chemical network onto a learned low-dimensional manifold, and, closest to this work, Branca & Pallottini (2024) train DeepONet emulators on KROME (Grassi et al. 2014) integrations to reproduce the *time evolution* of a nine-species ISM photochemistry network at a single point, reaching mean relative errors below 3 per cent with a ∼128× speed-up over the stiff ODE solver. All of these emulators are zero-dimensional: they map a local initial state to a local final state and treat every cell independently. The present work addresses a complementary question that a zero-dimensional emulator cannot: how should a per-cell surrogate represent the *non-local* information — principally UV shielding by surrounding gas (cf. Glover & Mac Low 2007; Glover et al. 2010) — that couples a cell's equilibrium $H_2$ density to its three-dimensional environment,

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and how well do the resulting models generalise to radiation-field strengths absent from training?

Three properties of the problem shape the answer. First, $n_{H_2}$ in our simulation suite spans roughly 16 orders of magnitude (from $\sim 10^{-12}$ to $\sim 10^4$ cm$^{-3}$), so all modelling is done in $\log_{10}$ space. Second, the atomic-to-molecular transition is sharp: the equilibrium abundance changes by orders of magnitude across a small range of shielding column. Third, and most importantly, a cell's chemical state is not a function of its local primitive variables alone; whether it is shielded from dissociating UV depends on the column of gas and dust between it and the radiation source.

A volumetric convolutional neural network (CNN) is the textbook answer to the non-locality problem: a 3D U-Net (Ronneberger et al. 2015; Çiçek et al. 2016) processing the whole volume can in principle learn shielding-like aggregations through its receptive field. But when the training set contains only a handful of simulation cubes at discrete parameter values, the volumetric sample count is tiny compared with the parameter count of even a small U-Net, and the behaviour of such models on out-of-distribution parameter values is not well characterised. We compare this learned approach against a deliberately simple engineered alternative: precomputed multi-scale box-filter averages of all input fields, appended to the per-cell feature vector of tabular models (gradient-boosted trees and multilayer perceptrons).

This paper makes four contributions. (i) We define a stringent out-of-distribution evaluation protocol — leave-one-$G_0$-out cross-validation over a controlled seven-simulation UV-parameter sweep — and show that it reverses the inflated conclusions that random cell-level splits would give. (ii) We show that multi-scale spatial neighbourhood features deliver a consistent accuracy gain to tabular models at negligible cost, and that a Ridge-stacked ensemble of gradient-boosted trees and a wide MLP reaches a mean log-space $R^2$ of 0.988. (iii) Because one input feature — the $H_2$ self-shielding factor $f_{H_2}$ — is itself produced by the chemistry solver, we quantify the surrogate's dependence on it with a dedicated ablation, finding that the baseline models lose only $\approx 0.04$ in mean $R^2$ when it is removed, that the spatial features recover about half of the loss, and that the fully solver-independent ensemble retains $R^2 = 0.977$. (iv) We characterise when the volumetric approach is and is not competitive: the best 3D U-Net configuration matches the tabular models on interpolation folds but is less robust at the extrapolation boundaries, far more sensitive to training configuration, and roughly two orders of magnitude more expensive to train.

The paper is structured as follows. Section 2 describes the data. Section 3 defines the evaluation protocol and reproducibility measures. Section 4 describes the feature engineering and models. Section 5 presents the main comparison, the $f_{H_2}$ ablation, and the error analysis of the final model. Section 6 reports supplementary experiments on cross-$G_0$ transfer and within-cube sampling geometry. Sections 7 and 8 give the discussion and conclusions.

## 2 DATA

### 2.1 Simulation suite

The data set consists of seven three-dimensional uniform-grid simulations from a UV-only chemistry suite. Each simulation occupies a $128 \times 128 \times 128$ grid of 2,097,152 cells, evolving identical turbulent ISM initial conditions under a spatially uniform, time-steady UV radiation field of strength $G_0$ in Habing units (Habing 1968). The seven runs differ in their imposed $G_0$ alone, sharing the same initial turbulent realisation and grid geometry; the suite is a controlled UV-parameter sweep, not a set of independent realisations. The evolved snapshots nonetheless differ in their hydrodynamic state as well as their chemistry — the median gas temperature rises from $\approx 1.0 \times 10^3$ K at $G_0 = 0.1$ to $\approx 1.1 \times 10^4$ K at $G_0 = 6.4$ as UV heating strengthens, with corresponding shifts in the density and velocity fields — while the large-scale structure remains strongly correlated across the suite. The $G_0$ values span a 64-fold range on a geometric sequence, $G_0 \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4\}$, and the snapshots represent quasi-equilibrium chemical states. The UV field illuminates each cube along the $+z$ axis, which breaks the full octahedral grid symmetry and dictates the augmentation strategy of Section 4.2.3. The dominant chemical processes in the suite are $H_2$ formation on grain surfaces (Cazaux & Tielens 2004) and Lyman–Werner photodissociation (Draine & Bertoldi 1996) attenuated by $H_2$ self-shielding computed along the illumination axis (Wolcott-Green et al. 2011). The total data set contains 14,680,064 cells.

We use this suite as a controlled testbed for surrogate methodology: the single-parameter design isolates the question of generalisation across the UV field strength from all other sources of variation. The corresponding limitation — that generalisation across turbulent realisations, metallicities, or density regimes is untested — is discussed in Section 7.4. A complete description of the simulation code, chemical network, and initial conditions (box size, resolution, driving, metallicity, dust model) is beyond the scope of this methods-focused paper and must accompany any future application of the surrogate to scientific inference; we state explicitly which conclusions depend only on the internal structure of the data (all of them) and which depend on the physical realism of the simulations (none directly).

**Table 1.** The 15 input features. "Transform" is the preprocessing applied before modelling. "Type" distinguishes local primitive variables, the global run parameter, and non-local quantities. "Deployable" indicates whether the feature would be directly available in a target application where the chemistry solver has not been run; the self-shielding factor $f_{H_2}$ is the single problematic input and is treated by ablation in Section 5.2.

| Feature | Units | Transform | Type | Deployable |
|---|---|---|---|---|
| $n_H$ | cm$^{-3}$ | $\log_{10}$ | local | yes |
| $T$ | K | $\log_{10}$ | local | yes |
| $n_{H^+}$ | cm$^{-3}$ | $\log_{10}$ | local | yes |
| `ext` | – | none | non-local proxy | yes |
| $f_{H_2}$ | – | $\log_{10}$ | non-local, solver | no |
| $G_0$ | Habing | $\log_{10}$ | global | yes |
| $v_x, v_y, v_z$ | code | none | local | yes |
| $B$ (6 face comps.) | code | none | local | yes |

### 2.2 Physical fields

Each cell carries 18 raw fields: three integer grid coordinates (excluded from model input), the target $n_{H_2}$, and 15 input features summarised in Table 1. Fields spanning many orders of magnitude are log-transformed,

$$x_\ell = \log_{10}(x + \varepsilon), \qquad \varepsilon = 10^{-30}, \tag{1}$$

where $\varepsilon$ guards against $\log(0)$ without affecting any physically realised value.

The $H_2$ self-shielding factor $f_{H_2}$ merits a precise statement. It is the attenuation factor applied to the Lyman–Werner photodissociation rate, computed by the chemistry solver from the $H_2$ column between each cell and the UV source. It is *not* algebraically derived from the cell's own $n_{H_2}$: we verified directly in the data that $f_{H_2}$ does not equal the local molecular fraction $2n_{H_2}/(n_H + 2n_{H_2})$ for any cell, and that cells with nearly identical local $n_{\mathrm{H_2}}$ carry $f_{\mathrm{H_2}}$ values differing by orders of magnitude depending on their depth within the cloud. It is nonetheless an output of the very calculation the surrogate aims to replace: in a deployment scenario where the solver has not been run, $f_{\mathrm{H_2}}$ would have to be estimated, e.g. from a dust-extinction proxy or an approximate radiative-transfer sweep. Its contribution to surrogate accuracy is therefore quantified separately (Section 5.2) rather than assumed. The dust-extinction proxy `ext` is also a column-type quantity, but unlike $f_{\mathrm{H_2}}$ it is computable from the density field alone without chemical information.

### 2.3 Target distribution

Figure 1 shows the distribution of $\log_{10}(n_{\mathrm{H_2}})$ across $G_0$. The target spans ≈16 orders of magnitude, from $\sim 10^{-12}\,\mathrm{cm^{-3}}$ in UV-exposed gas to $\sim 10^{4}\,\mathrm{cm^{-3}}$ in shielded cores, making linear-space regression numerically useless and motivating log-space modelling throughout. At $G_0 = 0.1$ the distribution is bimodal — a UV-exposed population near $\log_{10}(n_{\mathrm{H_2}}) \approx -10$ and a self-shielded molecular population near $\log_{10}(n_{\mathrm{H_2}}) \approx 0$ — and the per-cube mean falls from $-3.1$ at $G_0 = 0.1$ to $-7.2$ at $G_0 = 6.4$ as photodissociation becomes dominant. This strong dependence of the marginal target distribution on $G_0$ is what makes leave-one-$G_0$-out evaluation demanding: each held-out cube has a target distribution shifted relative to everything seen in training.

## 3 EVALUATION PROTOCOL

### 3.1 Leave-one-$G_0$-out cross-validation

All headline results use leave-one-$G_0$-out seven-fold cross-validation: each fold withholds one entire cube (≈2.1 M cells) and trains on the remaining six (≈12.6 M cells). Random cell-level splits would place cells from every cube — hence every $G_0$ and the single shared turbulence realisation — in both training and test sets; under that protocol a pointwise gradient-boosted tree already exceeds $R^2 = 0.99$, telling us nothing about generalisation. The intended application is prediction at a UV field strength that has not been simulated, and leave-one-$G_0$-out measures exactly that.

The two boundary folds require genuine extrapolation: no training data exist below $G_0 = 0.1$ or above $G_0 = 6.4$. The five interior folds require interpolation between bracketing cubes. Empirically, which folds are hardest depends on the model family (Table 2). Gradient-boosted trees lose most accuracy at the upward fold ($G_0 = 6.4$), where the dissociation-dominated equilibrium lies beyond the training range and tree leaves extrapolate as constants; the 3D U-Net degrades at both boundaries, most strongly at the downward fold ($G_0 = 0.1$), where the self-shielded molecular phase is most developed. The MLP and the ensembles built on it are instead weakest at the interior fold $G_0 = 0.8$, showing that $G_0$-extrapolation is not the only failure mode the protocol exposes.

### 3.2 Metrics

All primary metrics are computed in $\log_{10}(n_{\mathrm{H_2}})$ space: the coefficient of determination

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{2}$$

the root-mean-square error (RMSE), and the mean absolute error (MAE), the latter two in dex. As a secondary metric we report a clipped linear-space $R^2$: predictions are clipped to within 1 dex of the true range of the fold before exponentiation, preventing a handful of outlier cells from dominating the linear-space statistic. Linear-space $R^2$ is reported for completeness only; with a 16-dex target range it is dominated by the few densest cells.

### 3.3 Reproducibility

Every experiment is logged to a timestamped JSON file containing the full run configuration (feature list, hyperparameters, random seed, software environment) and per-fold metrics. Random seeds are fixed per fold (a global seed plus the fold index). The data files are fingerprinted by a SHA-256 checksum recorded in each log, so any change to the input data is detectable. Every result quoted in this paper is traceable to an archived log, and the complete pipeline runs end-to-end on one consumer GPU.

## 4 METHODS

### 4.1 Multi-scale spatial neighbourhood features

The dominant non-local effect on $n_{\mathrm{H_2}}$ is UV attenuation by surrounding gas. To give pointwise models access to spatial context without training a volumetric network, we precompute uniform box-filter averages of all 15 input fields. For each feature $f$ and cube, the tabular values are reshaped to a $128^3$ volume and filtered at kernel widths $k \in \{3, 5, 7\}$ voxels:

$$\bar{f}_{\boldsymbol{i},k} = \frac{1}{k^3} \sum_{\boldsymbol{j} \in \mathcal{N}_k(\boldsymbol{i})} f_{\boldsymbol{j}}, \tag{3}$$

where $\mathcal{N}_k(\boldsymbol{i})$ is the $k \times k \times k$ neighbourhood centred on $\boldsymbol{i}$ (kernel volumes of 27, 125, and 343 voxels). The box filter is separable, so the cost is $O(N)$ per scale independent of $k$; computing all 45 filtered fields for all seven cubes takes ≈30 s. The filtered values are concatenated with the 15 local features into a 60-dimensional vector:

$$\boldsymbol{x}_i^{(60)} = \left[ \boldsymbol{x}_i^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,3}^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,5}^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,7}^{(15)} \right]. \tag{4}$$

Figure 2 contrasts this representation with per-cell modelling. Spatial averages of the deployable column-type fields (density, extinction) act as crude shielding estimators; this is why the spatial features partially substitute for $f_{\mathrm{H_2}}$ in the ablation of Section 5.2.

### 4.2 Models

#### *4.2.1 Gradient-boosted trees*

We use XGBoost (Chen & Guestrin 2016) with the GPU histogram method: 400 boosting rounds, maximum depth 6, learning rate 0.1, row subsampling 0.3, column subsampling 0.8. Depth variants 4 and 8 change the mean $R^2$ by less than 0.003 and were retired. Features are standardised per training fold (a no-op for trees in principle; retained for pipeline uniformity).

#### *4.2.2 Wide MLP*

The multilayer perceptron has hidden widths $[512, 512, 256, 128]$ with batch normalisation (Ioffe & Szegedy 2015) and ReLU activations, trained with Adam (Kingma & Ba 2015) (learning rate $10^{-3}$, weight decay $10^{-5}$), batch size 262,144, 100 epochs per fold, cosine-annealed learning rate, and mixed precision. The full training fold is preloaded to GPU memory and batched by on-device permutation.

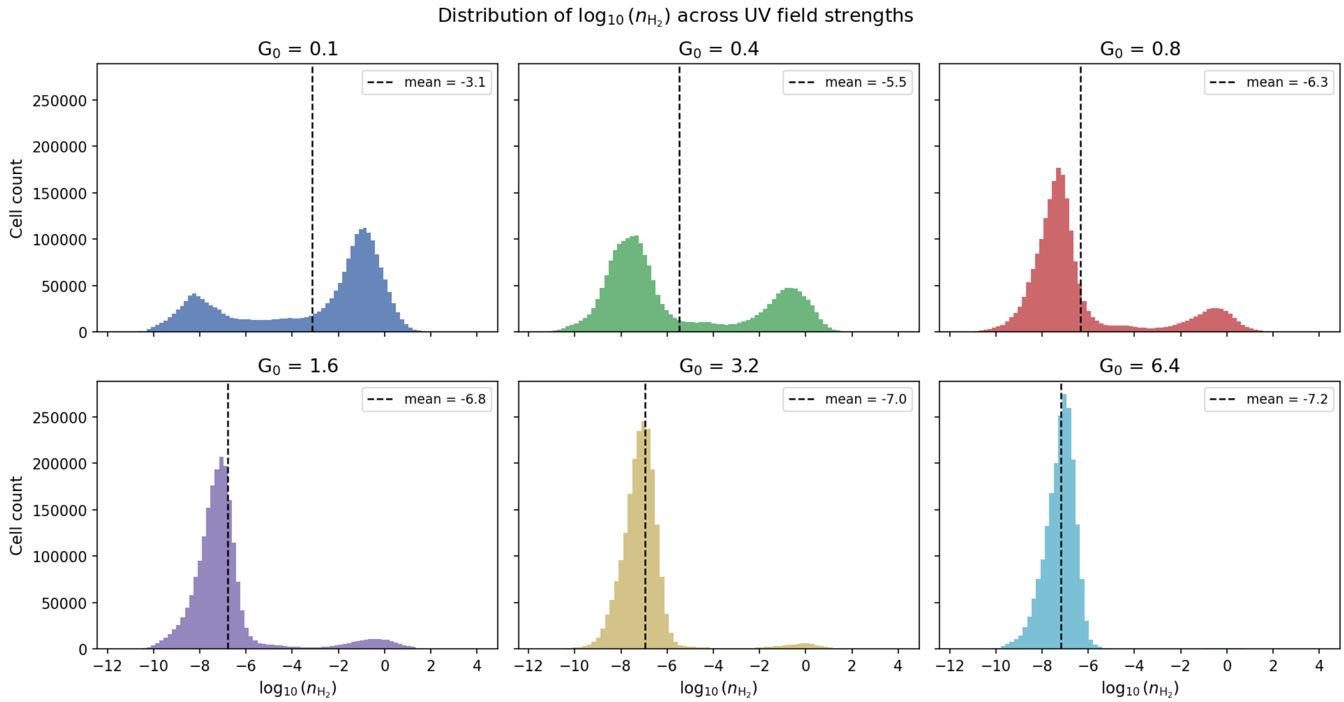


**Figure 1.** Distribution of $\log_{10}(n_{H_2})$ across six UV field strengths. The seventh cube, $G_0 = 0.2$, closely resembles $G_0 = 0.1$ and is omitted from this figure only to avoid visual redundancy; it is included in all quantitative analyses. The bimodal structure at low $G_0$ (UV-exposed and self-shielded populations) collapses to a single UV-dominated mode as $G_0$ increases. Dashed lines mark per-cube means. All modelling is performed in log-space.

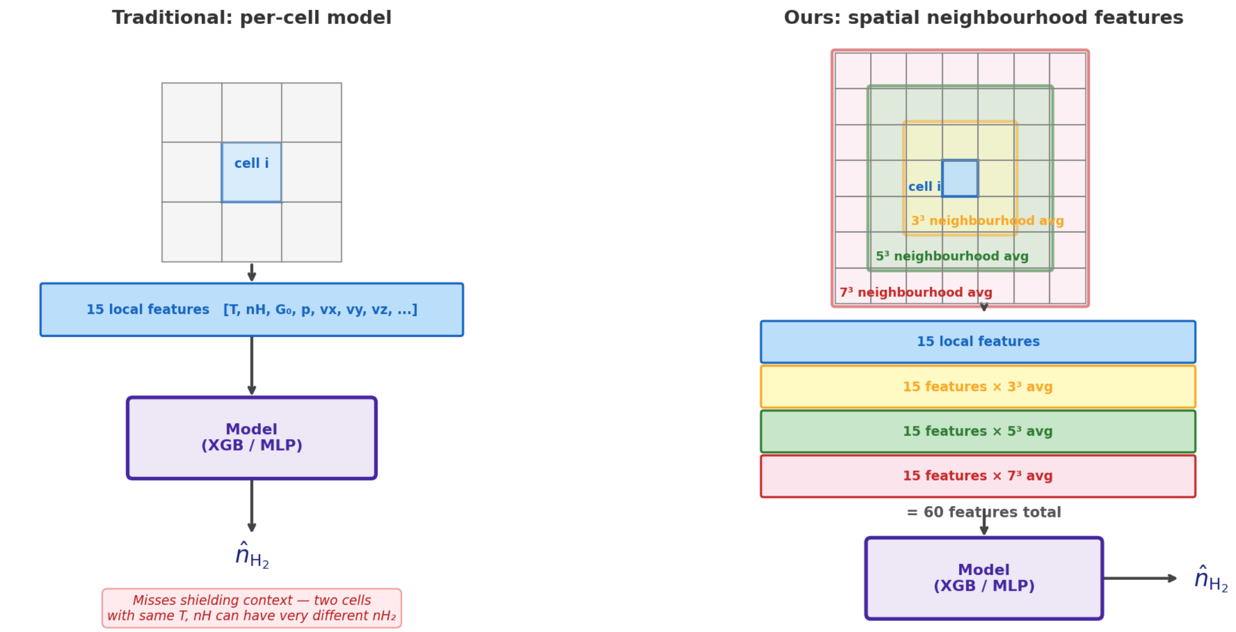


**Figure 2.** Schematic comparison of per-cell modelling (*left*) and the multi-scale spatial neighbourhood representation used here (*right*). Note that the 15-feature baseline already includes one non-local quantity, the solver-computed self-shielding factor $f_{H_2}$; what the spatial features add is explicit neighbourhood context for *all* fields, including the directly observable ones, at three nested scales.

Predictions are clipped to the training-target range ±2 dex. Narrower and residual-block (He et al. 2016) variants give mean $R^2$ within ~0.02 with no consistent ordering across runs; differences between MLP architectures are comparable to seed-to-seed variation.

#### 4.2.3 3D U-Net

The volumetric baseline is a 3D U-Net (Ronneberger et al. 2015; Çiçek et al. 2016) with three encoder stages, a bottleneck spanning the full (downsampled) volume, trilinear-upsampling decoder, and skip connections. Because each training sample is one entire volume (batch size 1), instance normalisation (Ulyanov et al. 2017) replaces batch normalisation. Inputs are average-pooled from $128^3$ to $64^3$, an 8× memory reduction; the tabular models, by contrast, use all $128^3$ cells, so the CNN comparison is not at equal resolution (Section 5.3). Channel inputs and the training target are standardised per fold from training-cube statistics; per-fold target standardisation is necessary for stable training because the target mean shifts by several dex from fold to fold (Section 2.3) and otherwise dominates the loss. Training uses Adam (learning rate $5 \times 10^{-4}$, weight decay $10^{-5}$), gradient-norm clipping at 1.0, and checkpoint selection on held-out volume loss. L2 weight decay is the sole regulariser: a residual variant that adds dropout (Srivastava et al. 2014) of 0.1 at reduced capacity underperforms the dropout-free baseline at matched epochs (mean $R^2$ 0.948 versus 0.974), with the loss concentrated at the two extrapolation folds (0.867 and 0.815 versus 0.900 and 0.945); we did not isolate the dropout contribution from the architecture change. Capacity variants with 16, 32, and 64 base channels (1.5 M, 5.8 M, and 23 M parameters) are compared; 150 epochs per fold.

Data augmentation respects the broken symmetry of the illuminated cube: of the 48 octahedral operations, only the eight $z$-preserving ones (rotations about $z$ and reflections through planes containing $z$) are physically valid. Scalars transform by index permutation, velocities as polar vectors, and the face-centred magnetic field components as axial vectors with the appropriate face swaps. Using all 48 operations injects unphysical samples and degrades validation accuracy. Each fold therefore trains on $6 \times 8 = 48$ augmented volumes.

### 4.3 Ensembles

Two ensembles combine the spatial-feature XGBoost and MLP. The *equal-weight ensemble* averages their per-cell predictions. The *Ridge-stacked ensemble* fits a Ridge regression ($\alpha = 1$) meta-learner on out-of-fold predictions: for each held-out fold, the meta-learner is trained on the base-model predictions for the six training cubes (each obtained from folds where that cube was itself held out, so no cell's own-fold prediction enters its meta-training) and applied to the held-out cube. The two base models have complementary inductive biases — piecewise-constant, conservative extrapolation for the trees; smooth interpolation for the MLP — and the stacked combination is the best model in every experiment (Section 5.1).

## 5 RESULTS

### 5.1 Main comparison

Table 2 reports the leave-one-$G_0$-out performance of all models on the full 15-feature input set (plus 45 spatial features where indicated). Spatial features improve every tabular model: XGBoost from 0.950 to 0.963 and the MLP from 0.954 to 0.968 in mean $R^2$, with the largest gains concentrated in the low-$G_0$ folds where shielded structure is most developed. The Ridge-stacked spatial ensemble is the best model at $R^2 = 0.988 \pm 0.013$ (RMSE 0.25 dex), and outperforms the equal-weight ensemble (0.978) in this and every archived repetition of the comparison; with only seven folds the advantage is modest but consistent. The hardest fold for the tree-based models is the upward extrapolation boundary $G_0 = 6.4$ ($R^2 = 0.883$ for XGBoost), barely helped by spatial features (0.885) but fully compensated in the stacked ensemble (0.999), which can lean on the MLP's smooth extrapolation where the trees saturate.

### 5.2 Dependence on the self-shielding factor

Because $f_{H_2}$ is produced by the solver the surrogate aims to replace, Table 3 repeats the comparison with $f_{H_2}$ excluded from both the local features and the spatial averages (14 local + 42 spatial features). The baseline XGBoost loses 0.039 in mean $R^2$ ($0.950 \to 0.911$) and the wide MLP loses 0.045 ($0.954 \to 0.909$), confirming that $f_{H_2}$ carries substantial information not recoverable from the remaining local features, while also showing that the surrogate degrades gracefully, not catastrophically, when restricted to solver-independent inputs: a deployment that estimates shielding only approximately (or not at all) retains a usable model. The spatial-feature variants recover roughly half of the gap, as expected if neighbourhood averages of density and extinction act as crude column estimators: the ablation penalty shrinks from 0.039 to 0.022 for XGBoost ($0.963 \to 0.941$) and from 0.045 to 0.016 for the MLP ($0.968 \to 0.953$), and the fully solver-independent stacked ensemble reaches $R^2 = 0.977 \pm 0.017$, within 0.011 of its with-$f_{H_2}$ counterpart. For the volumetric baseline, the available evidence is coarser: a U-Net trained without the density, ionisation, extinction, *and* shielding channels (11 inputs) falls from 0.974 to 0.874, but this removes four features at once and is reported only as an upper bound on the $f_{H_2}$ effect for the CNN.

### 5.3 The volumetric baseline

The best U-Net configuration (32 base channels, 150 epochs, weight decay only) reaches $R^2 = 0.974 \pm 0.035$: indistinguishable from the tabular spatial models on the five interior folds (all $R^2 \geq 0.993$) but weaker at both extrapolation boundaries (0.900 at $G_0 = 0.1$, 0.945 at $G_0 = 6.4$, versus 0.969/0.999 for the stacked ensemble). Two further differences matter in practice. First, *configuration sensitivity*: across capacity variants and repeated training runs, U-Net mean $R^2$ ranged from 0.80 to 0.97 (16-channel variant 0.824, 64-channel 0.801–0.889, 32-channel 0.878–0.974 depending on training protocol), whereas XGBoost results are bit-identical across runs and MLP variation stays within ~0.01. With six training volumes (48 after augmentation) and 1.5–23 M parameters, the volumetric models sit in a regime where training noise is large. Second, *cost*: a full seven-fold U-Net comparison takes hours on an RTX 3090, versus minutes for the tabular pipeline including spatial-feature construction. We did not pursue CNN-specific remedies (deeper augmentation, pretraining, equal-resolution training at $128^3$) and therefore make no claim about the ultimate capability of volumetric architectures; the operational conclusion is that on a suite of this size the engineered spatial features achieve equal or better accuracy with far greater robustness and lower cost.

**Table 2.** Leave-one-$G_0$-out log-space $R^2$ for all evaluated models on the full feature set. "Feat." gives the input dimension (15 local; 60 = 15 local + 45 spatial; vol. = volumetric input at $64^3$). The best result in each column is bold. Tabular results are from the master comparison run (Section 3.3); the 3D U-Net row is its best training configuration (32 base channels, 150 epochs, no dropout). The U-Net operates on 2× downsampled volumes, so its comparison to the tabular rows is not at equal resolution.

| Model | Feat. | Mean $R^2$ | $\sigma$ | $G_0$=0.1 | $G_0$=0.2 | $G_0$=0.4 | $G_0$=0.8 | $G_0$=1.6 | $G_0$=3.2 | $G_0$=6.4 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ridge-stacked ensemble (spatial) | 60 | **0.988** | **0.013** | 0.969 | **0.993** | **0.995** | 0.965 | **0.997** | **0.998** | **0.999** |
| Equal-weight ensemble (spatial) | 60 | 0.978 | 0.016 | **0.988** | 0.991 | 0.993 | 0.945 | 0.975 | 0.988 | 0.968 |
| Wide MLP + spatial | 60 | 0.968 | 0.041 | 0.979 | 0.986 | 0.988 | 0.871 | 0.963 | 0.996 | 0.996 |
| XGBoost + spatial | 60 | 0.963 | 0.033 | 0.976 | 0.982 | 0.985 | 0.982 | 0.975 | 0.955 | 0.885 |
| Wide MLP | 15 | 0.954 | 0.038 | 0.943 | 0.957 | 0.911 | 0.890 | 0.988 | 0.992 | 0.993 |
| XGBoost | 15 | 0.950 | 0.029 | 0.956 | 0.948 | 0.972 | **0.972** | 0.969 | 0.951 | 0.883 |
| 3D U-Net (32 ch., best config.) | vol. | 0.974 | 0.035 | 0.900 | 0.993 | 0.995 | **0.997** | 0.996 | 0.993 | 0.945 |

**Table 3.** Ablation of the solver-derived self-shielding factor $f_{H_2}$. Each cell gives the mean ± fold std. of log-space $R^2$ over the seven leave-one-$G_0$-out folds. "With" rows use all 15 local features (60 with spatial); "without" rows exclude $f_{H_2}$ from the local features and from the spatial averages (14 local; 56 with spatial). All runs use the identical pipeline, seeds, and hyperparameters.

| Feature set | XGBoost | Wide MLP | XGB + spatial | MLP + spatial | Stacked ensemble |
|---|---|---|---|---|---|
| With $f_{H_2}$ | 0.950 ± 0.029 | 0.954 ± 0.038 | 0.963 ± 0.033 | 0.968 ± 0.041 | 0.988 ± 0.013 |
| Without $f_{H_2}$ | 0.911 ± 0.033 | 0.909 ± 0.057 | 0.941 ± 0.029 | 0.953 ± 0.027 | 0.977 ± 0.017 |

### 5.4 Error analysis of the final model

For per-cell error analysis we use a dedicated prediction run of the Ridge-stacked spatial ensemble that saves full $128^3$ prediction volumes for each held-out fold (per-fold $R^2$ = 0.979, 0.984, 0.982, 0.954, 0.978, 0.981, 0.915 for $G_0$ = 0.1–6.4; mean 0.968 ± 0.023). This run uses a shorter MLP schedule (30 epochs) for the inner stacking stage than the master comparison, and is accordingly slightly more conservative than the headline row of Table 2; we use it because the saved volumes permit the spatially resolved analysis below. Figures 3–8 are all derived from these predictions.

Figure 4 shows predicted versus true $\log_{10}(n_{H_2})$ for all seven folds: the model reproduces the bimodal structure in every fold, with scatter concentrated in the diffuse UV-exposed population and a tight locus along the identity for the shielded population. Figure 5 shows the residual distributions. These are narrow (core widths 0.04–0.45 dex, shrinking monotonically with $G_0$) but *not* centred at zero: the run carries a systematic positive bias of +0.2 to +0.5 dex at the interior folds, largest at $G_0$ = 0.8, meaning the model over-predicts $n_{H_2}$ by a factor of ~2–3 for typical cells in those folds while still ordering cells accurately (high $R^2$). The bias is a property of this prediction run that ensemble recalibration would largely remove; we report it rather than hide it, and note that fold-level $R^2$ alone would not have revealed it.

Figure 6 stratifies the errors by gas density. The largest mean absolute errors occur in the *low*-density deciles of the low-$G_0$ folds (up to ~0.8 dex in decile 3 of $G_0$ = 0.1), where small shielding-column differences move cells across the steep atomic-to-molecular transition. Conversely, restricting the evaluation to the densest cells *lowers* $R^2$ (panels a and d) even though absolute errors there are small: within the shielded cores the target variance is small, so $R^2$ — a variance-normalised metric — is a harsh statistic in the high-density tail. Both views are shown to avoid the misleading impression either one gives alone. Figure 7 compares predicted and true marginal distributions per fold; the bimodal shape and peak locations are reproduced, with the positive bias visible as a slight rightward shift of the predicted distribution at interior folds.

Figure 8 shows mid-plane slices of truth, prediction, and absolute error. At low $G_0$ the field is rich in filamentary shielded structure and the residuals concentrate at cloud boundaries; at high $G_0$ the field is nearly uniform and residuals are small and unstructured.

## 6 SUPPLEMENTARY EXPERIMENTS

### 6.1 Single-cube transfer across $G_0$

To measure how far the learned mapping transfers across the UV parameter, we train the stacked spatial ensemble on each individual cube and predict all seven, producing a 7 × 7 matrix of log-space $R^2$ (Figure 9). In-sample (diagonal) fits average $R^2$ = 0.997. Transfer degrades systematically with $G_0$ separation: averaged over the matrix, one geometric step (×2 in $G_0$) gives $R^2$ = 0.966, two steps 0.888, three steps 0.729, four steps 0.452, five steps 0.158, and the full 64-fold separation 0.006, with substantial spread between directions and cube pairs at the larger separations. The mean off-diagonal $R^2$ is 0.706. The learned local mapping is thus approximately universal within a factor of ~2–4 in $G_0$ and degrades beyond it, which (i) validates leave-one-$G_0$-out as a non-trivial protocol — each held-out cube is one step from its nearest training neighbour, so single-cube knowledge alone would cap performance near 0.97, below what the six-cube training sets achieve — and (ii) confirms that none of the seven $G_0$ values is redundant in the training set.

### 6.2 Within-cube sampling geometry

We next ask how reconstruction of a *single* cube's $n_{H_2}$ field depends on which cells are used for training — a question about sampling geometry in the simulation volume, not (we emphasise) a claim about observational survey design, since real observations deliver projected, beam-convolved, noisy quantities rather than simulation fields, and $f_{H_2}$ is not an observable. For each cube we train the stacked spatial ensemble on a subset of cells and evaluate on the complement, comparing uniformly random subsets (1–75 per cent), contiguous half-cube slabs along each axis, and compact cubic sub-regions (1–50 per cent).

In these experiments the spatial features must respect the partial coverage. Field volumes are masked before filtering: cells outside the training subset are set to zero, and the box filter of equation (3) is applied to the masked volume,

$$\bar{f}^{\mathrm{obs}}_{\boldsymbol{i},k} = \frac{1}{k^3} \sum_{\boldsymbol{j} \in N_k(\boldsymbol{i})} m_{\boldsymbol{j}} f_{\boldsymbol{j}}, \qquad m_{\boldsymbol{j}} = \begin{cases} 1 & \boldsymbol{j} \in \text{training subset}, \\ 0 & \text{otherwise}, \end{cases} \tag{5}$$

with *no* renormalisation by the local sample count $\sum_{\boldsymbol{j}} m_{\boldsymbol{j}}$: a neigh-

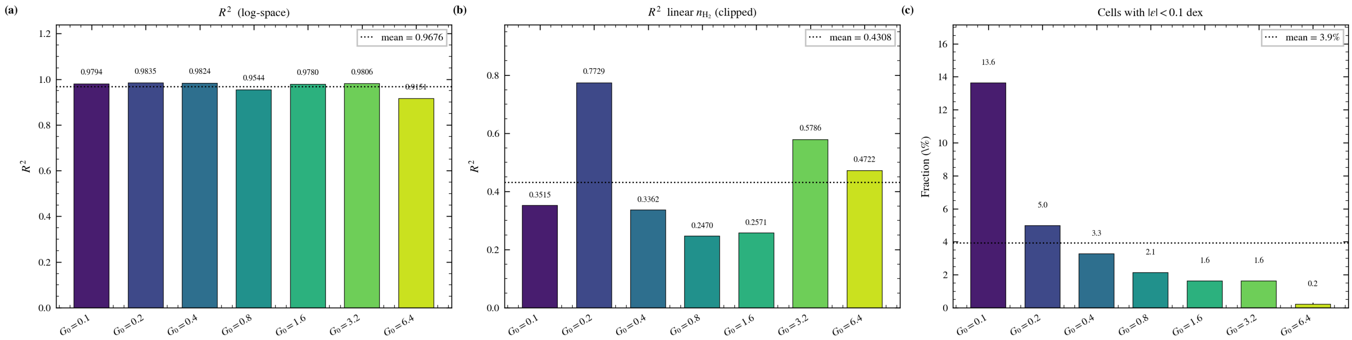


**Figure 3.** Summary of the final-model prediction run. *(a)* Log-space $R^2$ per held-out fold; the upward extrapolation fold $G_0 = 6.4$ is the hardest for this model, followed by $G_0 = 0.8$; the dashed line marks the seven-fold mean. *(b)* Clipped linear-space $R^2$ (secondary metric; Section 3.2); the much lower values reflect the dominance of the densest cells in linear space. *(c)* Fraction of cells with absolute residual below 0.1 dex, highest at low $G_0$ where the well-predicted shielded population is largest.

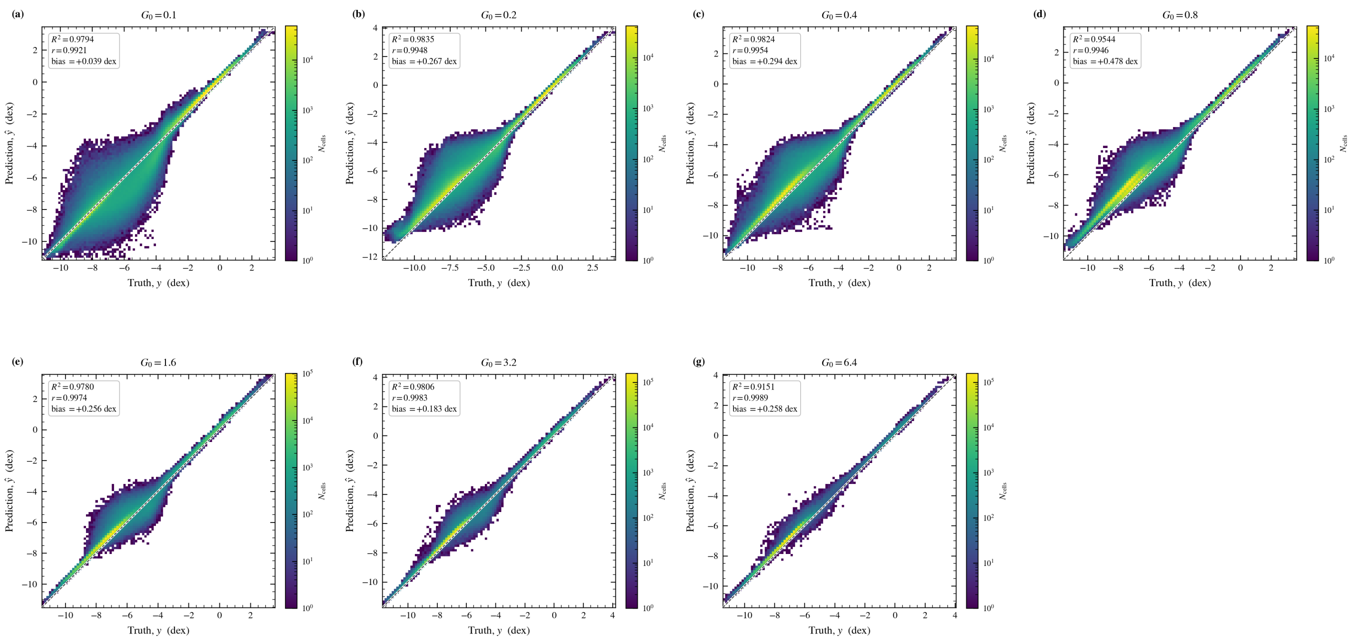


**Figure 4.** Predicted versus true $\log_{10}(n_{H_2})$ per held-out fold (two-dimensional histograms, log colour scale; dashed line = identity). Per-fold $R^2$, RMSE, and MAE are annotated. The bimodal target structure is reproduced in every fold; scatter is dominated by the diffuse UV-exposed population.

bourhood with no sampled cells yields zero, and a sparsely sampled neighbourhood yields its sampled-cell sum diluted by the full kernel volume. The local (unfiltered) features of test cells use their true values; only the target labels and the neighbourhood context are restricted to the training subset. This convention treats the surrounding field values as unknown wherever they were not sampled, and its dilution property matters for interpreting the sparse-coverage results below.

Figure 10 and the run statistics give a consistent picture across all seven cubes. Uniformly random coverage of 10 per cent or more reconstructs the field accurately (test $R^2$ mean over cubes: 0.977 at 10 per cent, 0.982 at 25, 0.986 at 50); 5 per cent gives 0.902. At 1 per cent random coverage the expected number of sampled cells inside even a $7^3$ kernel is only $\sim$3.4, the diluted spatial features carry little signal, and the outcome is unstable: mean $R^2$ of +0.24 with a range of −1.74 to +0.90 across cubes. Contiguous geometries show the opposite failure mode: a half-cube slab — 50 per cent of the volume — fails on every axis (mean $R^2$ of −0.65, −0.25, −0.25 for $x$, $y$, $z$ slabs, with minima as low as −5.1), because the unsampled half contains physical regimes entirely absent from training. Compact boxes are intermediate and nearly flat in coverage (mean $R^2 \approx 0.76$–0.86 from 1 to 50 per cent). Notably, at 1 per cent coverage a compact box *outperforms* random sampling (0.76 versus 0.24): inside the box the masked spatial features are exact, whereas under sparse random sampling they are uniformly diluted. Geometry, not volume, is the controlling variable, and the best geometry depends on the coverage budget: contiguous regions win when coverage is very sparse, uniform sampling wins everywhere above a few per cent.

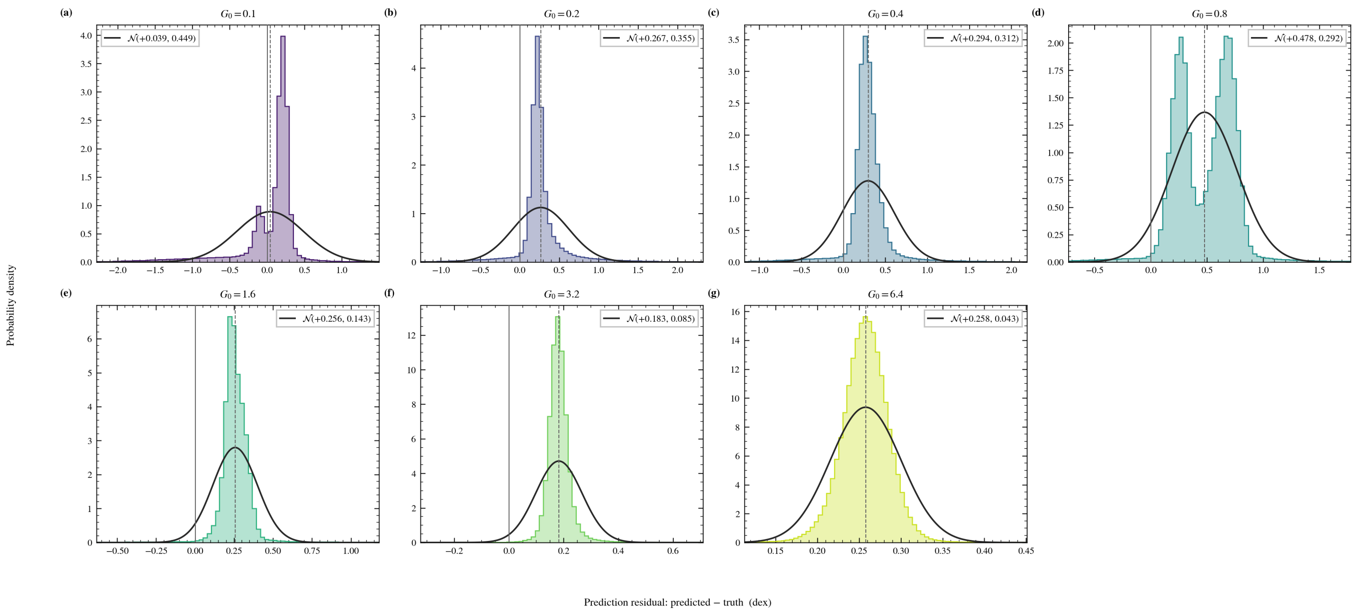


**Figure 5.** Residual distributions (predicted − true, dex) per fold, with best-fit normal curves and their $(\mu, \sigma)$ annotated. Distribution widths decrease monotonically with $G_0$. Note the systematic positive bias at the interior folds (up to +0.5 dex at $G_0 = 0.8$) and the multi-modal structure at several folds: the Gaussian curves summarise location and scale only and should not be read as a claim of normality.

## 7 DISCUSSION

### 7.1 Feature engineering versus architecture

On this suite, input representation dominates architecture choice among the tabular models. The spatial features add +0.013 and +0.015 mean $R^2$ to XGBoost and the MLP respectively — several times larger than any architecture variation we tested (tree depth, MLP width and depth, residual connections, all $\lesssim 0.01$ with no consistent ordering) — and stacking adds a further +0.02 over the best single model. This is consistent with the broader observation that tree-based models with appropriate features remain highly competitive on tabular problems (Grinsztajn et al. 2022). The box-filter means are a deliberately minimal spatial representation; anisotropic kernels aligned with the illumination axis, or cumulative sums along it (a discrete column integral), are natural refinements that we have not exhausted. A physics-based reference point — an analytic molecular-fraction prescription (e.g. Krumholz et al. 2009; Gnedin & Kravtsov 2011; Bialy & Sternberg 2016) evaluated cell-by-cell on the suite — would usefully anchor the ML numbers; published formulae predict equilibrium molecular fractions from column-averaged quantities under geometric assumptions the cell-level task does not satisfy, so a fair comparison requires a calibration step that we leave to future work.

### 7.2 Volumetric versus per-cell representations

The 3D U-Net is not categorically worse here — at its best configuration it matches the tabular models on every interpolation fold — but it is less robust in two specific senses: its accuracy falls off at both extrapolation boundaries, and its mean performance varies by ∼0.2 in $R^2$ across capacity and training configurations, a variance the tabular models simply do not exhibit. With 48 augmented training volumes against millions of parameters, this is the expected signature of an under-constrained estimator, and we caution against reading the comparison as a general statement about volumetric networks: at equal input resolution, with more independent simulations, or with pretraining, the ranking could change. What the comparison does establish is a practical default for limited-data suites: engineered neighbourhood features capture the dominant spatial signal at a small fraction of the cost, and they do so deterministically.

### 7.3 Deployability and the shielding factor

The ablation of Section 5.2 addresses the most delicate methodological question in this study: whether the surrogate's accuracy is borrowed from the solver through $f_{\mathrm{H_2}}$. The answer is quantitative. $f_{\mathrm{H_2}}$ is genuinely informative (worth 0.039–0.045 in mean $R^2$ to the baseline models) but not load-bearing: both baseline models retain $R^2 \approx 0.91$ without it, and the spatial-feature ensemble retains 0.977, because neighbourhood averages of the deployable column-type fields substitute for much of the shielding information. Deployment therefore has a spectrum of options: with a cheap approximate shielding estimate (e.g. from extinction), accuracy should fall between the two rows of Table 3; with no shielding estimate at all, the without-$f_{\mathrm{H_2}}$ row applies directly. What our data cannot quantify is the effect of using an *inconsistent* shielding estimator (one with different systematics from the solver's), which would require forward-modelling the estimator itself.

### 7.4 Limitations

*Dataset scope.* The seven simulations grow from one turbulent initial realisation and differ only in their imposed $G_0$, so their structures are strongly correlated even though the evolved fields are not identical (Section 2.1). All generalisation claims are therefore claims about the UV-field axis; transfer across turbulence realisations, metallicities, dust models, or resolutions is untested and plausibly harder. The suite is also quasi-equilibrium and uniformly illuminated; time-dependent or anisotropic radiation fields would invalidate the $z$-preserving augmentation and may change the role of the spatial features.

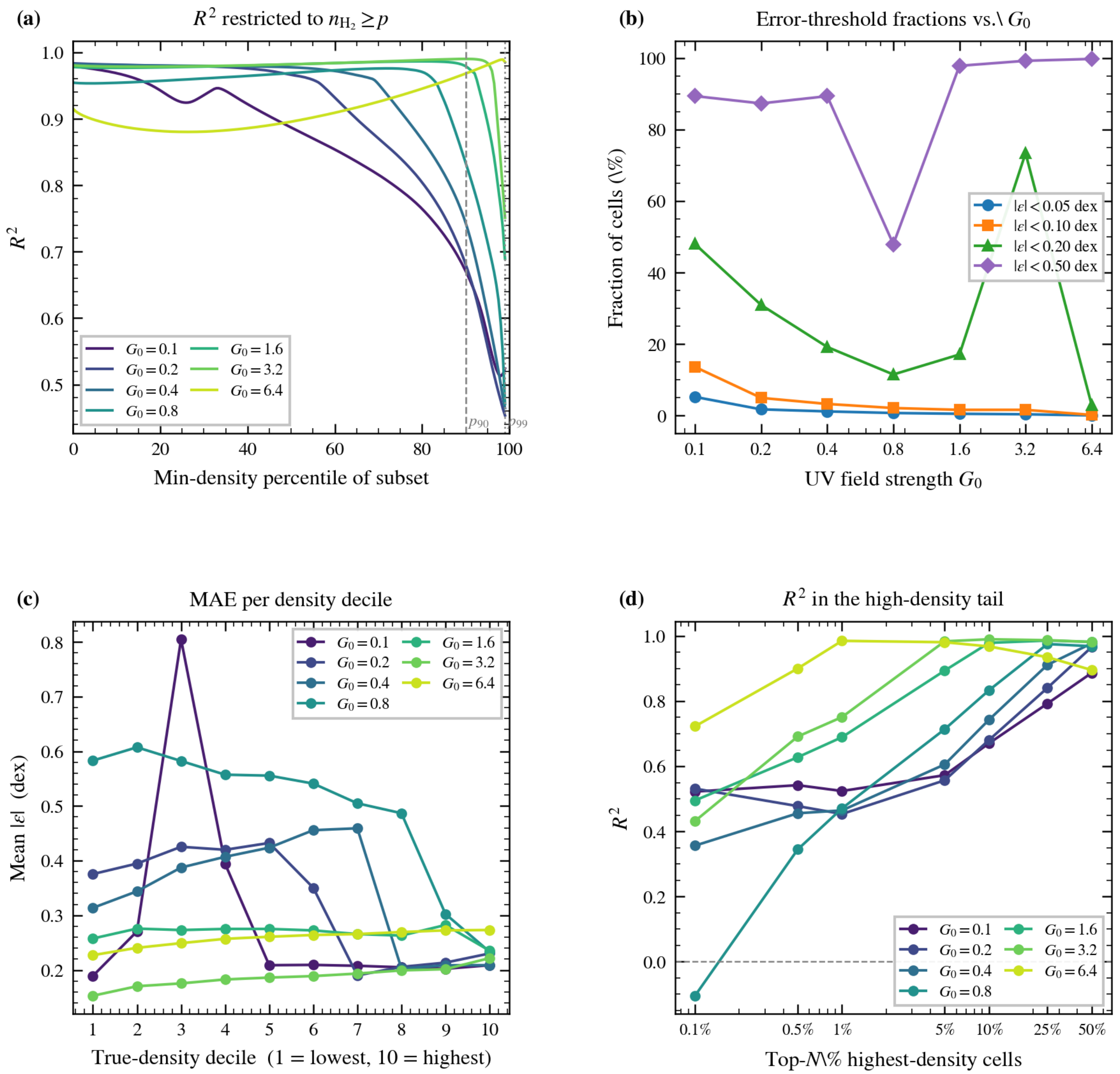


**Figure 6.** Density-stratified error analysis. *(a)* $R^2$ on the subset of cells above a minimum-density percentile: $R^2$ *decreases* toward the dense tail because the within-subset target variance shrinks faster than the errors. *(b)* Fraction of cells with absolute residual below fixed thresholds versus $G_0$; the dip at $G_0 = 0.8$ reflects that fold's +0.5 dex bias. *(c)* MAE per true-density decile: errors peak in the low-density deciles of the low-$G_0$ folds, at the steep atomic-to-molecular transition; high-$G_0$ folds are flat at ~0.2 dex. *(d)* $R^2$ within the top-$N$ per cent densest cells.

*Simulation provenance.* This paper characterises surrogate methodology on a controlled private suite, and its conclusions are internal to that suite (relative model performance, feature ablations, sampling geometry). Application of the trained surrogate to scientific inference requires the full simulation specification — code, chemical network, box size, resolution, metallicity, dust-to-gas ratio, driving — which is not part of this study.

*CNN comparison.* The U-Net was trained at 2× reduced resolution for memory reasons and was not exhaustively tuned; the configuration-sensitivity result is itself evidence that any single U-Net number should be read with a wide error bar.

*Final-model bias.* The per-cell analysis run carries a systematic positive bias of up to +0.5 dex at interior folds (Section 5.4). Applications sensitive to absolute calibration rather than ranking should recalibrate per-$G_0$ or include a bias-correction stage, which fold-level $R^2$ alone would not reveal.

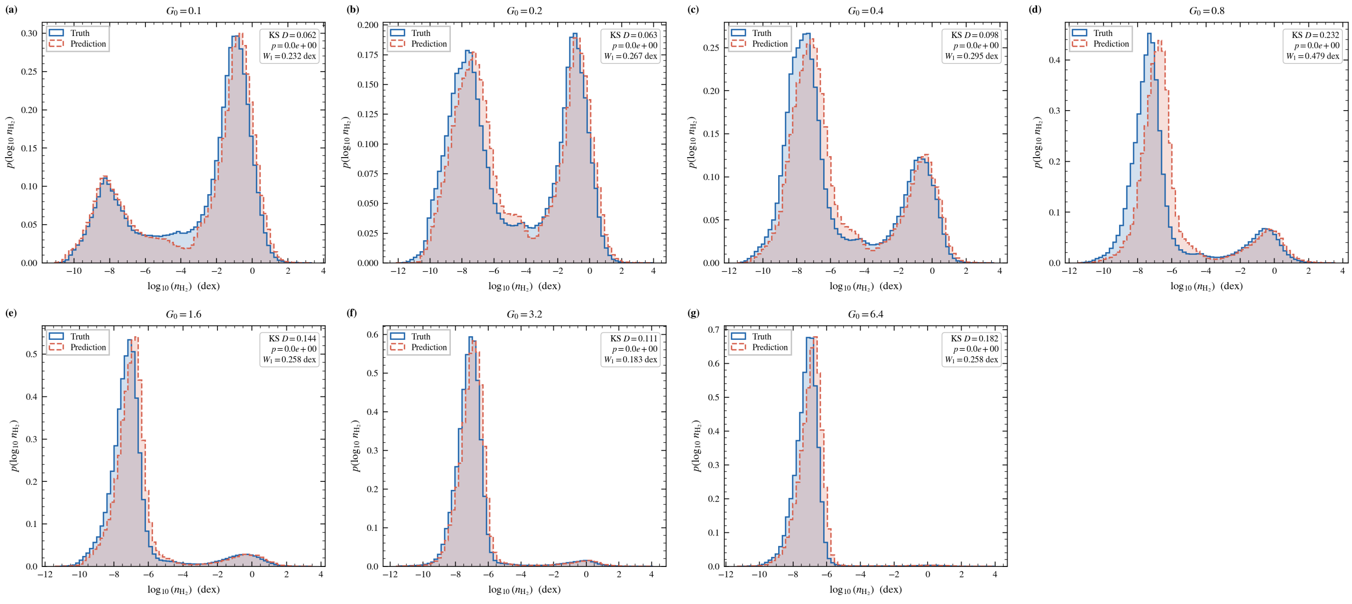


**Figure 7.** True (filled) versus predicted (outline) marginal distributions of $\log_{10}(n_{H_2})$ per fold, with $R^2$, KL divergence $D_{KL}(P_{true} \parallel P_{pred})$ (computed from equal-width histograms over the common support), and MAE annotated. The bimodal structure and peak positions are reproduced; the interior-fold positive bias appears as a small rightward shift of the predicted distribution.

## 8 CONCLUSIONS

We evaluated machine-learning surrogates for the molecular hydrogen number density in a controlled suite of seven $128^3$ UV-only ISM chemistry simulations under leave-one-$G_0$-out cross-validation. The main conclusions are:

(i) Multi-scale box-filter neighbourhood features (kernel widths 3, 5, 7 voxels; ≈30 s to compute for the full suite) improve every tabular model: XGBoost from $R^2 = 0.950$ to 0.963 and a wide MLP from 0.954 to 0.968. A Ridge-stacked ensemble of the two spatial-feature models reaches $R^2 = 0.988 \pm 0.013$ (RMSE 0.25 dex) and is the best model in every experiment.

(ii) The solver-derived self-shielding factor is highly informative but not indispensable: excluding it costs the baseline trees 0.039 in mean $R^2$ ($0.950 \to 0.911$) and the wide MLP 0.045 ($0.954 \to 0.909$); the spatial features halve the penalty, and the fully solver-independent stacked ensemble retains $R^2 = 0.977 \pm 0.017$ (versus 0.988 with $f_{H_2}$). The surrogate is therefore usable with solver-independent inputs alone, at a small and quantified accuracy cost.

(iii) The best 3D U-Net configuration ($R^2 = 0.974 \pm 0.035$) matches the tabular models on all interpolation folds but degrades at both extrapolation boundaries and varies by ~0.2 in mean $R^2$ across training configurations, at roughly two orders of magnitude higher training cost. On suites of this size, engineered spatial context is the more robust default; this is not a claim about volumetric architectures in data-rich regimes.

(iv) The learned local mapping transfers across a factor of ~2–4 in $G_0$ (single-cube transfer $R^2 = 0.966$ at one geometric step, 0.888 at two) and degrades beyond it, validating leave-one-$G_0$-out as a non-trivial protocol and confirming that none of the seven simulations is redundant.

(v) Within a cube, reconstruction quality is controlled by sampling geometry rather than sample volume: 10 per cent uniformly random coverage achieves $R^2 = 0.977$ while 50 per cent contiguous slabs fail ($R^2 < 0$), and at very sparse coverage (~1 per cent) compact contiguous regions outperform random sampling because the masked spatial features remain exact inside them. These are simulation-space results; translating them to observational survey design would require forward modelling of observables.

The full tabular pipeline — feature construction, training, and prediction for all seven folds — runs in minutes on a single consumer GPU, and the trained surrogate predicts all 2,097,152 cells of a new cube in seconds.

## ACKNOWLEDGEMENTS

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## SOFTWARE

Python 3.14.5; PyTorch 2.12 (Paszke et al. 2019); XGBoost 3.2 (Chen & Guestrin 2016); NumPy 2.4 (Harris et al. 2020); SciPy 1.17 (Virtanen et al. 2020); scikit-learn 1.8 (Pedregosa et al. 2011); Matplotlib 3.10 (Hunter 2007). All experiments were run on a single NVIDIA GeForce RTX 3090. Random seeds are fixed per fold; every experiment writes a timestamped JSON log containing its full configuration, per-fold metrics, software environment, and a SHA-256 fingerprint of the input data.

## DATA AVAILABILITY

All experiment logs, trained-model outputs, and prediction volumes are archived as timestamped JSON and `.npz` files and are available from the author on request. The simulation suite is not yet public; requests for data access may be directed to the corresponding author.

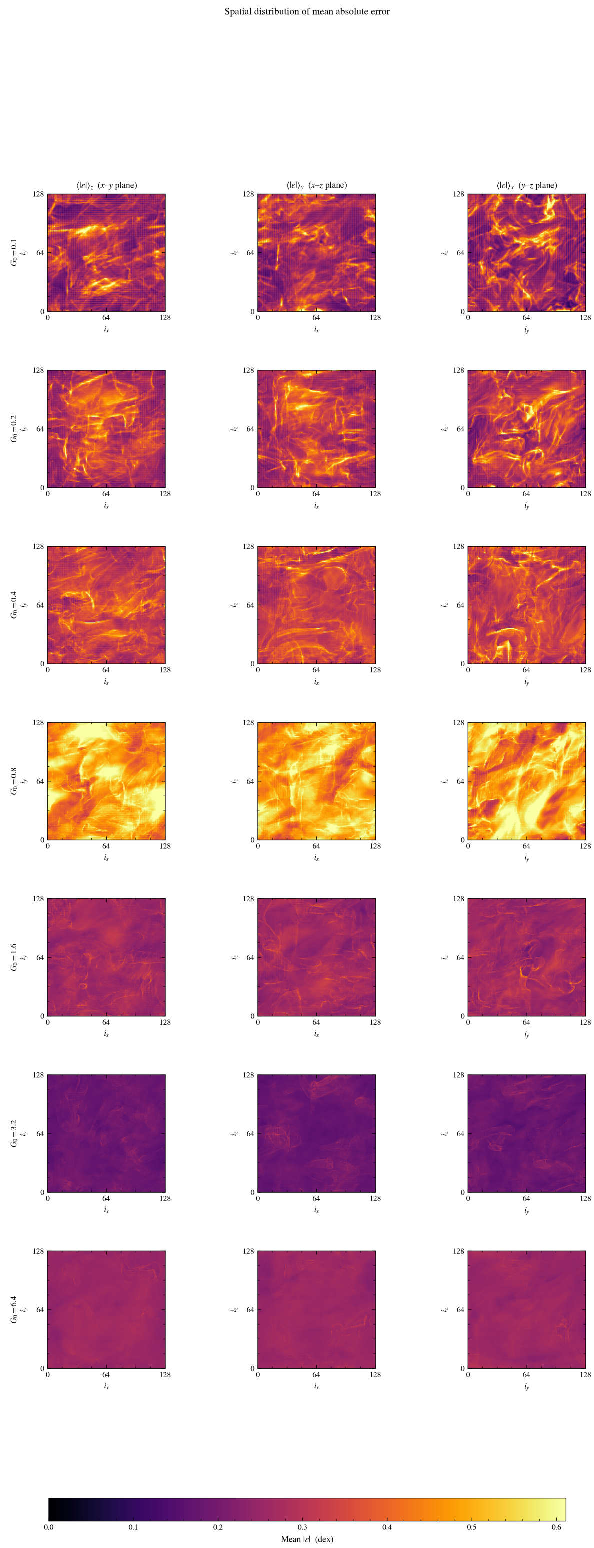


**Figure 8.** Mid-plane slices of the $\log_{10}(n_{H_2})$ field for each $G_0$ (rows): ground truth, prediction, and absolute error (columns). At low $G_0$ residuals trace the boundaries of shielded filaments; at high $G_0$ the field is nearly uniform. Colour scale is shared within each row.

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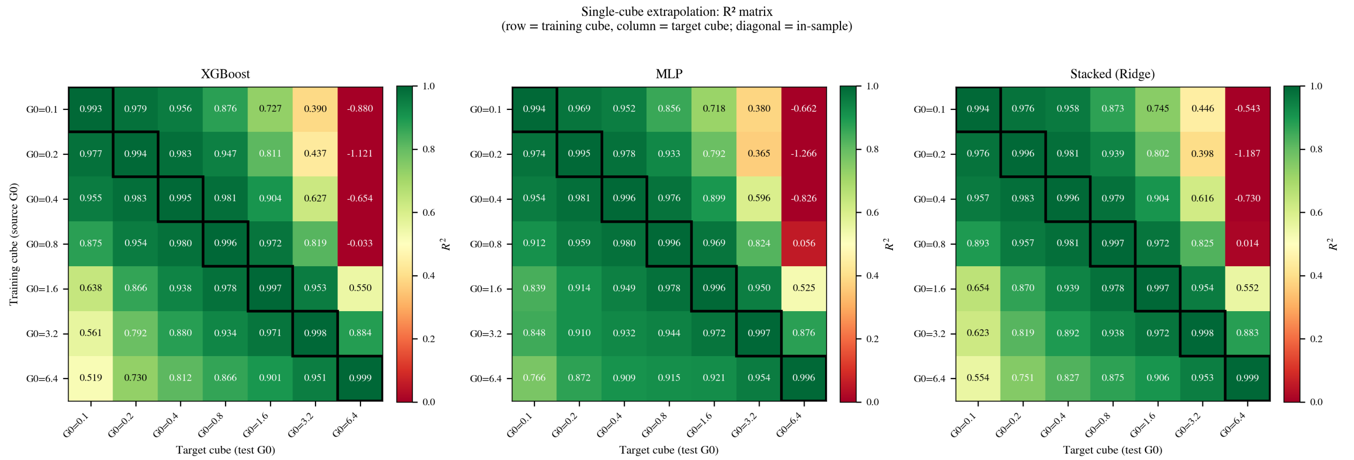


**Figure 9.** Single-cube transfer: log-space $R^2$ for models trained on one cube (rows) and evaluated on each cube (columns), for XGBoost (*left*), MLP (*centre*), and the stacked ensemble (*right*). Diagonal cells are in-sample fits. Transfer quality decays with $G_0$ separation; values for separations of four or more geometric steps are poor and can be strongly negative.

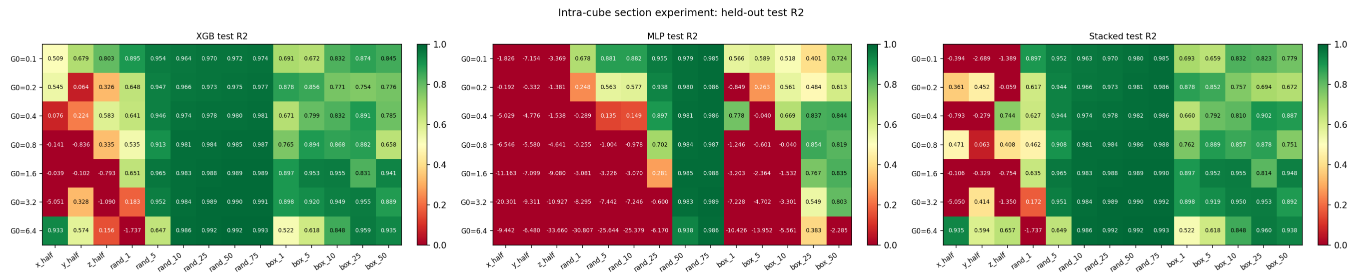


**Figure 10.** Within-cube sampling geometry: test $R^2$ per training geometry (columns) and cube (rows) for XGBoost (*left*), MLP (*centre*), and the stacked ensemble (*right*). Random coverage ≥5–10 per cent reconstructs the held-out cells well in every cube; 50 per cent contiguous slabs fail in every cube; compact boxes are intermediate; 1 per cent random coverage is unstable, varying from $R^2 \approx 0.9$ to strongly negative between cubes.


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